

Implementing mathematical formulations in Python

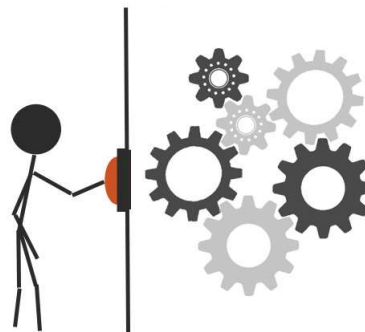
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Textbook: Postek, K. and Zocca, A. and Gromicho, J. and Kantor, J. (2025)
Hands-On Mathematical Optimization with Python, Cambridge
 University Press, doi 10.1017/9781009493512

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MODELLING TOLS:

- AMPL, GAMS, MPL,
- MATLAB, R
- Python, C, C#, Julia



SOLVERS (PLI):

Cbc
 COPT
 CPLEX
 FICO-Xpress
 Gurobi
 MINTO
 MOSEK
 OCTERACT
 proxy
 qsopt_ex
 scip
 SYMPHONY

More complete list: <https://neos-server.org/neos/solvers/index.html>

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Optimisation for Industry

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Python packages

- PuLP
- Pyomo
- Python-MIP

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Formulate the following production planning problem

The company BIM (Best International Machines) produces two types of microchips, logic chips (1g silicon, 1g plastic, 4g copper) and memory chips (1g germanium, 1g plastic, 2g copper). Each of the logic chips can be sold for a 12€ profit, and each of the memory chips for a 9€ profit. The current stock of raw materials is as follows: 1000g silicon, 1500g germanium, 1750g plastic, 4800g copper.

How many microchips of each type should be produced to maximize profit while respecting the availability of raw material stock?

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Formulation

Decision variables:

$x \geq 0$, # of logic chips

$y \geq 0$, # of memory chips

max $12x + 9y$

s.t. $x \leq 1000$ (silicon)

$y \leq 1500$ (germanium)

$x + y \leq 1750$ (plastic)

$4x + 2y \leq 4800$ (copper)

$x, y \geq 0$.

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Python implementation using Pyomo

```
import pyomo.environ as pyo
SOLVER = pyo.SolverFactory('glpk')
model = pyo.ConcreteModel("BIM production planning")
model.x1 = pyo.Var(domain=pyo.NonNegativeReals) # Decision variables and their
domains
model.x2 = pyo.Var(domain=pyo.NonNegativeReals)
model.profit = pyo.Objective(expr=12 * model.x1 + 9 * model.x2, sense=pyo.maximize) # Objective function
model.silicon = pyo.Constraint(expr=model.x1 <= 1000) # Constraints
model.germanium = pyo.Constraint(expr=model.x2 <= 1500)
model.plastic = pyo.Constraint(expr=model.x1 + model.x2 <= 1750)
model.copper = pyo.Constraint(expr=4 * model.x1 + 2 * model.x2 <= 4800)
SOLVER.solve(model) # Solve and print solution
print(f"x = ({model.x1.value:.1f}, {model.x2.value:.1f})")
print(f"optimal value = {pyo.value(model.profit):.1f}")
```

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BIM under uncertainty

- Suppose that there is uncertainty regarding the selling prices of the microchips. Instead of just the nominal prices 12 and 9, BIM estimates that prices may more generally take values, i.e., $\{(12,9), (11,10), (8,11)\}$.

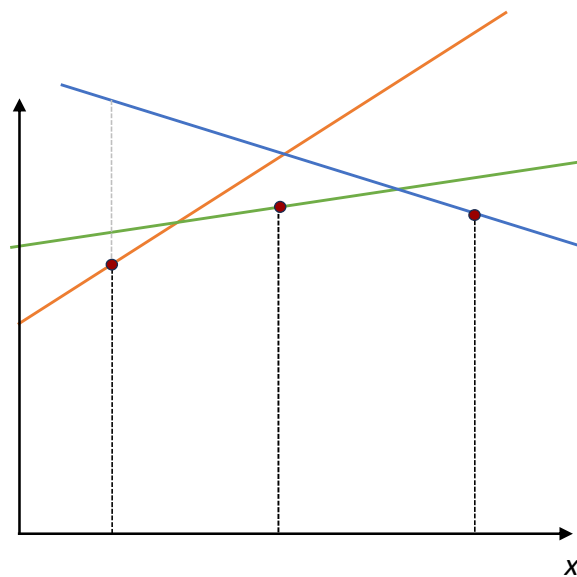
$$\max_{x,y} \min_{k \in K} c_1^k \cdot x + c_2^k \cdot y$$

$$A \begin{pmatrix} x \\ y \end{pmatrix} \leq b$$

$x \leq 1000$ (silicon)
 $y \leq 1500$ (germanium)
 $x + y \leq 1750$ (plastic)
 $4x + 2y \leq 4800$ (copper)
 $x, y \geq 0$

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Example



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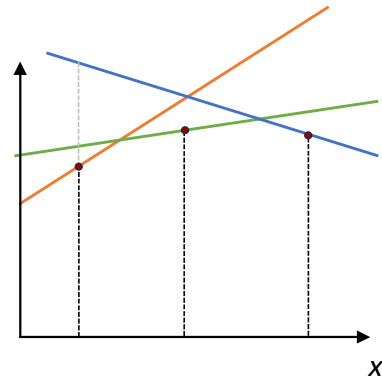
LP reformulation

$$\max_{x,y} \min_{k \in K} c_1^k \cdot x + c_2^k \cdot y$$

$$\max z$$

$$z \leq c_1^k \cdot x + c_2^k \cdot y \quad \forall k \in K$$

$$A \begin{pmatrix} x \\ y \end{pmatrix} \leq b$$



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Formulation ctd.

$$\max \quad z$$

$$\text{s.t.} \quad z \leq 12x + 9y$$

$$z \leq 11x + 10y$$

$$z \leq 8x + 11y$$

$$x \leq 1000 \quad (\text{silicon})$$

$$y \leq 1500 \quad (\text{germanium})$$

$$x + y \leq 1750 \quad (\text{plastic})$$

$$4x + 2y \leq 4800 \quad (\text{copper})$$

$$x, y \geq 0.$$

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Analysis

k	c_x	c_y	$cost$	x	y
1	12	9	17700	650	1100
2	11	10	18150	650	1100
3	8	11	18500	250	1500
$max\ min$			17700	583.3	1166.7

$solution$	c_x	c_y	$k=1$	$k=2$	$k=3$
1	650	1100	17700	18150	17300
2	650	1100	17700	18150	17300
3	250	1500	16500	17750	18500
$maxmin$	583.3	1166.7	17500	18083.3	17500

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Implement the formulation in python

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Exercise: Implement the diet problem